NOISEMAP · REGULATORY AND STANDARDS REGISTER

NoiseMap regulatory and standards register

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue (kit 1.8 pipeline)

NoiseMap regulatory and standards register

Which standards and regulations bear on a street sound level node, where, and what is needed. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Seven of fourteen items are marked to assess and none is a certification NoiseMap holds. NoiseMap is a research and educational prototype and a non-certified indicative instrument; it must not be presented as a class 1 or class 2 sound level meter or as legal evidence. Method: STANDARDS section 23. Dates are the day a link was found or read in the 2026-11-02 session; where a standard number is "as known" or "as cited earlier", the text was not re-read and that is said in the row.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Directive 2002/49/EC, assessment and management of environmental noise (Lden, Lnight, 4.0 m assessment height)	EU	Strategic noise maps and indicators	Gives the 4.0 m height and the indicators NoiseMap reports against; the node is not a mapping instrument under it	Design lead	Does not apply as a compliance duty; used as the measurement convention. Source: EUR-Lex, checked 2026-11-02
G2	IEC 61672-1:2013, sound level meters, specifications	International	Class 1 and class 2 sound level meters; weighting and tolerances	NoiseMap follows the definitions of A and C weighting where practical (PR-005) and must not be presented as a class 1 or class 2 instrument. Source: IEC webstore, cited in NSM-PRB-001 (2026-10-02)	Design lead	Does not apply as certification; certification needed later if a class is ever claimed (TRL 4 and later, on hold)
G3	ETSI EN 300 220 series, short-range devices 25 MHz to 1000 MHz; EU868 duty cycle of 1 %	EU	Radio parameters and duty cycle in the 868 MHz band	The LoRaWAN radio of the FieldNode core; airtime in NSM-CAL-001 F (PR-012). Number as cited in the earlier NSM-REQ-001 R12; text not re-read in this session	Design lead	Applies; met on paper for duty cycle (NSM-CAL-001 F)
G4	Radio Equipment Directive 2014/53/EU	EU	Placing radio equipment on the EU market (essential requirements, conformity, marking)	A node with its own radio is radio equipment; the FieldNode module's conformity may carry over only if the module is used within its approval. Source: EUR-Lex, https://eur-lex.europa.eu/eli/dir/2014/53/oj, checked 2026-11-02	Regulatory consultant	To assess; certification needed later for any product placed on the market
G5	The Things Network fair use policy (30 s uplink airtime per day per node)	Network (community)	Shared network airtime	Drives the interval rule at SF10 to SF12 (PR-012). Source: https://www.thethingsnetwork.org/forum/t/the-things-network-fair-use-policy/47689, checked 2026-11-02 (policy page listed by search; the 30 s figure is as cited in NSM-REQ-001 R12)	Design lead	Applies; met on paper (NSM-CAL-001 F)
G6	Regulation (EU) 2016/679, General Data	EU	Personal data; a recording of speech could be personal data	NoiseMap stores and sends levels only, by design (PR-001); whether a level time series at a street can ever identify a	Data protection adviser	To assess with a data protection adviser before

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	Protection Regulation			person is to assess. Source: EUR-Lex, https://eur-lex.europa.eu/eli/reg/2016/679/oj, checked 2026-11-02		any deployment; other jurisdictions to assess
G7	Low Voltage Directive 2014/35/EU and EMC Directive 2014/30/EU	EU	Electrical safety and electromagnetic compatibility of equipment	Option A runs from a 3.2 V cell and a 6 W panel, which is below the Low Voltage Directive's mains range, so only EMC is expected to apply; option C has mains inside. Directive numbers as known; links not checked in this session	Regulatory consultant	To assess (A: EMC; C: both)
G8	Battery rules for lithium cells: UN transport tests (UN 38.3), IEC 62133-2 cell safety, Regulation (EU) 2023/1542	International, EU	Safe lithium cells, transport and battery placing on the market	The 19 Wh LiFePO4 cell is a FieldNode core part; the cell maker's test reports are the evidence. Numbers as known; texts not checked in this session	Regulatory consultant	To assess (FieldNode core item; cited once there)
G9	IEC 60529, degrees of protection (IP code)	International	Ingress protection ratings	PR-008 asks for IP65; no rating is claimed for the printed head, only a design intent (drip skirt, membrane, bored windscreen). Number as known	Design lead	Applies as a target; certification needed later
G10	Pole and column loading, for example the EN 40 series for lighting columns	EU, national	Loads a column is designed to carry	The node adds about 96 N at 3.5 to 4 m in a 35 m/s gust (NSM-CAL-001 H); the pole owner confirms the pole can take it. Which national rule applies is to assess	Pole owner	To assess per jurisdiction and pole
G11	ANSI C136.41 dimming receptacle and Zhaga Book 18 (outdoor luminaire sensor and communication nodes)	US (ANSI), international (Zhaga)	The receptacle on a luminaire that a sensing node plugs into	Only option C. Zhaga Book 18 edition 3.0 (April 2021) adds a hybrid with an ANSI C136.41 receptacle. Source: https://zhagastandard.org/books/overview/smart-interface-between-outdoor-luminaires-and-sensing-communication-modules-18.html, checked 2026-11-02	Lighting operator	To assess (option C only)
G12	IEC 62368-1, safety of audio/video,	International	Product safety including	Only option C (mains converter inside a sealed puck); not a	Regulatory consultant	To assess (option C only);

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	information and communication equipment		isolation and creepage	STEM row. Number as known; text not checked	and electrician	certification needed later
G13	Work at height and traffic management at installation (national rules)	National	Fall protection, lifts, traffic control	The safety notes in NSM-PRC-001; installation is by trained crews with the pole owner's permission	Installer	Applies at installation; local rules to assess
G14	Noise exposure limits for people using a calibrator (NIOSH recommended 85 dBA for 8 h, 3 dB exchange)	US guidance, widely used	Hearing protection time at 94 dB	The 94 dB calibrator check is about 2 minutes (STM S19, PR-009). Source: https://www.cdc.gov/niosh/bulletin/2022/noise-50.html, checked 2026-11-02	Technician	Applies as guidance; met by design

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- European Parliament and Council. 2002. Directive 2002/49/EC relating to the assessment and management of environmental noise. https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32002L0049 (cited in NSM-PRB-001).
- International Electrotechnical Commission. 2013. IEC 61672-1, Electroacoustics: Sound level meters, Part 1. https://webstore.iec.ch/en/publication/5708 (cited in NSM-PRB-001).
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- Zhaga Consortium. Book 18, smart interface between outdoor luminaires and sensing and communication modules. https://zhagastandard.org/books/overview/smart-interface-between-outdoor-luminaires-and-sensing-communication-modules-18.html (read 2026-11-02).
- National Institute for Occupational Safety and Health. NIOSH Noise: a 50-year timeline. https://www.cdc.gov/niosh/bulletin/2022/noise-50.html (found 2026-11-02).